

The storyline, and the math under it

Namyar et al., “End-to-End Performance Analysis of Learning-enabled Systems,” *HotNets* ’24.

The whole paper is one chain of forced moves. Each section exists because the previous one left exactly one option open. Hold the chain and the paper is yours.

Question. How badly can a deployed learning-enabled system underperform — not on its test set, but on the worst input that exists?

1. Why the DNN alone is the wrong object

DOTe’s DNN emits split ratios; the operator cares about MLU. These are not the same thing: Figure 3 shows two different split-ratio vectors producing *identical* MLU, and the same vector producing different MLU under different demands. So a bound on the DNN’s output error implies nothing about system performance.

Forced: analyze $\mathcal{M}_{adv}(H(x))$ — an objective over the whole pipeline $H = \text{DNN} + \text{post-processor} + \text{routing} + \text{MLU}$.

2. Why the existing two families both fail

White-box (MetaOpt, Virelay) encodes the system as one MILP/SMT instance. Two blockers: it needs piecewise-*linear* activations (the authors had to swap DOTe’s activation out just to run the comparison), and jointly encoding components makes one component’s output another’s variable — non-convex, non-linear, unsolvable at scale. It found nothing in 6 hours.

Black-box random search uses no structure, so it wanders. It found 1.22x.

Forced: use *partial* information — enough to get a direction, not enough to need a global model. That is the gradient.

3. Why a gradient is available at all

This is the paper’s enabling fact, and its most compressed sentence: *DNNs are piecewise sub-differentiable*.

Piecewise. Fixing every ReLU’s on/off state fixes a region of input space, and on that region the network is affine, $H(x) = W_{\text{eff}}x + b_{\text{eff}}$. The regions are convex polyhedra; the gradient is exact and constant inside each.

Sub-differentiable. On a region boundary there is no derivative, but there is a non-empty set of slopes of lines touching the graph and staying below it:

$$\partial\sigma(0) = \{ g : \sigma(z) \geq \sigma(0) + gz \ \forall z \} = [0, 1]$$

Any element is a subgradient; autodiff picks one. (Strictly this is the *Clarke* subdifferential — the convex-analysis definition above needs convex f , and DNNs are not convex. Same $[0, 1]$, weaker assumptions.)

Why that is enough. Write the system as a composition and apply the chain rule:

$$\nabla_x \mathcal{M}_{adv} = \nabla_y \mathcal{M}_{adv} \cdot \nabla_z H_2 \cdot \nabla_x H_1$$

Each H_i is differentiated **in place** — analytically, or estimated from samples of its input/output behavior. No component is ever written as a closed-form expression for a solver. *That* is the gray-box saving: modeling one component at a time is easy; modeling their joint dependencies is what killed white-box.

4. Why the objective is a ratio, and why that hurts

Absolute MLU is not a measure of failure — scale every demand up and any system congests. The meaningful quantity is distance from optimal:

$$\mathcal{M}_{adv}(d) \triangleq \frac{\text{MLU}_{DOTE}(d)}{\text{MLU}_{OPT}(d)} = \max_f \frac{\text{MLU}_{DOTE}(d)}{\text{MLU}(d, f)} \quad (2)$$

Why this is non-convex in d , twice over: the numerator is a non-convex DNN composed with a max (ReLU is convex, but the next layer’s weights may be negative, and convexity survives composition only under a non-decreasing outer function); and a quotient of convex functions is not convex anyway. Gradient ascent on it reaches a local optimum only.

5. Why the denominator can simply be deleted

For a fixed routing f , every link load is linear in d , so

$$\text{MLU}(\kappa d, f) = \kappa \text{MLU}(d, f) \implies \text{MLU}_{OPT}(\kappa d) = \kappa \text{MLU}_{OPT}(d)$$

Positive homogeneity. So **every ray of demands crosses the surface $\text{MLU}_{OPT} = 1$ exactly once**, at $\kappa = 1/\text{MLU}_{OPT}(d)$. That surface is a cross-section of the demand cone: one representative per *direction*, magnitude discarded. Restrict to it and the denominator is identically 1:

$$\mathcal{M}_{adv}(d) \triangleq \text{MLU}_{DOTE}(d), \quad d \in \{d \mid \exists f : \text{MLU}(d, f) = 1\} \quad (3)$$

The hidden assumption. This is lossless only if DOTE’s split ratios are unchanged by rescaling — then MLU_{DOTE} is homogeneous too and the ratio is constant along each ray. A DNN has no reason to be scale-invariant, and the authors only *condition* on it (“if DOTE’s split ratios remain the same”). The fallback is to re-run the whole search over $\{d \mid \exists f : \text{OPT}(d, f) = P\}$ for several P and keep the best — restoring the magnitude dimension. For MLU, $P = 1$ suffices.

6. Why a Lagrangian, and what λ actually does

Eq. (3) is constrained; gradient ascent needs an unconstrained objective. Fold the constraint in as a penalty:

$$\min_{\lambda} \max_{d, f} \mathcal{L}_{final} \triangleq \mathcal{M}_{adv}(d) + \lambda(\text{MLU}(d, f) - 1) \quad (4)$$

The sign of λ is the whole trick. The multiplier step is $\lambda \leftarrow \lambda - \alpha_{\lambda}(\text{MLU}(d, f) - 1)$, so congestion above 1 drives λ negative. Since $\nabla_f \mathcal{L} = \lambda \nabla_f \text{MLU}(d, f)$, a negative λ turns **ascent on f into descent on MLU** — which is exactly what the optimal routing must do. The formal $\max_f$ produces a *minimizing* routing. Solved by multi-step gradient descent-ascent: T ascent steps on (d, f) , one descent step on λ (Eq. 5).

7. What the search actually returns

A demand matrix that the optimal routes with **zero congestion**, together with the routing f that proves it, on which DOTE congests a link **6x** over. The gap is therefore attributable to DOTE’s decisions, not to a hard input — and Figure 5 shows those demands look nothing like its training distribution. 50 seconds, against 6 fruitless hours for MetaOpt.

The mental model, in one paragraph

You cannot model the system, so don't: differentiate it. Piecewise sub-differentiability makes every component locally linear, the chain rule glues those local linearizations into an end-to-end direction, and you walk the input uphill. Everything else is bookkeeping to make “uphill” well-posed — a ratio so the metric means something, homogeneity to kill the denominator, a multiplier to kill the constraint. The price is a local optimum and a single example.

Where it is thin: the chain rule for subdifferentials is only an inclusion, $\partial(f \circ g) \subseteq \partial f \cdot \partial g$; “estimate the gradient from samples” carries real weight in the pitch but is never exercised, since DOTE’s pipeline happened to be differentiable throughout; and the scale-invariance assumption in §5 is asserted conditionally, never established.